

EPI2ME AAV QC

Component verification

Executed evidence, explicit assumptions and a reproducible metadata investigation.

Contract checks	Exploratory probes	Harness unit tests
8 / 8 passed	4 recorded	28 passed

Purpose and scope

AAV preparation QC summarises reads associated with transgene, host and plasmid references. This experiment checks whether selected counts, denominators and sample labels are handled correctly by Oxford Nanopore's actual contamination-summary component.

Small synthetic alignment tables were supplied directly to the upstream CLI. NumPy, pandas and native SeqKit executed normally. The source was preserved unchanged. No full Nextflow run, sequencing, alignment or variant calling was performed.

Method

Expected results were declared before execution. Eight checks covered a known mixture, repeated alignments, row order, all-mapped and transgene-only inputs, sample identity, a missing column and a missing FASTA. Each case retains inputs, expectations, commands, exit status, diagnostics and any output.

Manual calculation: reads and alignments

Measure	C01: baseline	C02: extra alignment
Total input reads	100	100
Alignment rows	95	96
Unique mapped reads	95	95
Mapped / unmapped	95% / 5%	95% / 5%
Transgene alignments	80	81
Transgene / all alignment rows	80/95 = 84.21%	81/96 = 84.38%

The duplicate alignment changes the alignment denominator, not the count of mapped reads. Reference-category percentages cannot be interpreted as percentages of all input reads. Rounding can make category percentages sum to 100.01%.

Recorded environment

Windows; Python 3.12.14; NumPy 2.2.6; pandas 2.2.3; SeqKit 2.13.0. Target: wf-aav-qc v1.3.1, commit 43a4266fc30a131c9e2b49654a97a3fd59e41d94. Dependencies describe this experiment, not the full upstream container.

When a successful exit is not enough

OBS-002: inconsistent total-read metadata

Probe X03 supplied 95 distinct mapped reads but declared only 94 total input reads. The unmodified component exited with code 0 and produced:

Output	Observed value
Mapped	95 reads / 101.06382978723404%
Unmapped	-1 read / -1.0638297872340425%

The component subtracts the distinct mapped count from the supplied total without a local consistency guard. Proposed controls are to reject inconsistent metadata with a clear diagnostic and independently check nonnegative counts and read percentages between 0 and 100.

This is confirmed component behaviour under deliberately inconsistent input. Full-workflow reachability is unproven: upstream processing may prevent the condition. The controls are proposed, not implemented. No upstream issue or patch has been submitted.

Other boundary observations

X01/X02 raised ZeroDivisionError for an empty alignment table or zero total-read count. X04 included an unlisted reference in the alignment denominator without a matching named category. Each needs contract and workflow-reachability review before defect classification.

Separate FASTQ validator benchmark

True positive	False negative	True negative	False positive
12	0	8	0

Positive means one malformed FASTQ file under the project's four-line A/C/G/T/N, Phred+33 dialect. Twenty authored challenge files yielded sensitivity 12/12, specificity 8/8 and coverage 20/20. These examples are related to the unit-test design; they are not external validation or EPI2ME/clinical accuracy.

Traceability and remaining work

Requirements link to the Nextflow process, Python component, cases and saved results. Technical risk notes record failure consequences, proposed controls and review status. samtools/bcftools exercises remain BLOCKED locally; a container-free Eddie installer is prepared. Full workflow execution and independent review remain pending. Later CI runs have separate evidence; the first Linux run failed file-tool checks.

Evidence and attribution

Repository: evidence/component-20260911-01; evidence/validator-20260911-01; docs/component-test-plan.md; docs/observation-002.md; docs/component-risk-review.md. Upstream code is Oxford Nanopore's work, supplied unchanged with its licence. Clinical validation and medical-device compliance are outside the demonstrated scope. Development assistance is documented in NOTICE.md.

Source: [Oxford Nanopore wf-aav-qc](#), pinned revision